

Llemy user manual

21-11-2025

1. Enter the API key you were provided



API Key Login

Lo
go
ut

Enter your OpenAI API key

A light gray rectangular input field for the API key. On the right side, there is a red square button with three white dots and a black eye icon.

Please enter your OpenAI API key to continue.

2. Enter a password. It should be easy to remember. This will not be stored but will be used to generate a unique ID that cannot be traced back to you (see next page). Please use the same password if you use the app more than once.



API Key Login

Lo
go
ut

Enter your OpenAI API key

.....



OpenAI API key set

Enter a password to received a unique identifier. This will not be stored. Please remember your password for future sessions, or make sure to write down all IDs in the questionnaire.



Please enter a password to continue



API Key Login ↗

Lo
go
ut

✓ OpenAI API key is already set

Enter a password to received a unique identifier. This will not be stored. Please remember your password for future sessions, or make sure to write down all IDs in the questionnaire.

.....



Unique ID to
keep for survey



✓ Identifier set: de1204b40c9a9044feee2f1974d3ad75bfe0d61d. Please write it down for the survey.

3. Review
informed consent
and tick the
button of your
choice



I have read and agree to the informed consent information and I want to participate in the Llemmy user research.

☐ Yes ☐ No



View informed consent details

Please answer the consent question to continue.



Home



Informed consent form



Instructions

MINERVA Project Selection

Select MINERVA Project:

Liver Lipid Metabolis... ▾

Example Questions

What is the scope of this map? Give me a brief ...

What is the overall scope of the disease map (m...

What are the inputs, regulators, and phenotypic



Unique identifier is already set: de1204b40c9a9044fdee2f1974d3ad75bfe0d61d. Please write it down for the survey.

I have read and agree to the informed consent information and I want to participate in the Llemy user research.



Yes



No



View informed consent details

4. Use the app!

Llemy

An Agentic System for MINERVA Map Exploration

Select a MINERVA map in the drop-down menu on the left, and ask questions about it.

You can:

- Type your own question in the input box below
- Try one of the example questions from the sidebar All questions are independant from each other, Llemy uses only the data from the selected map to answer your question.

Ask about biology, transporters, or metabolism...



This is a prototype application for research purposes only.

This work was developed in the context of the ONTOX project and VHP4Safety project.

Developped with Streamlit, LangChain and OpenAI (gpt-4.1-nano). License Apache 2.0.



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I have read and agree to the informed consent information and I want to participate in the Llemetry user research.

☒ Yes ☐ No



View informed consent details

4.1 Review instructions

Llemetry

An Agentic System for MINERVA Map Exploration

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4.2 Select your map of interest

Ask about biology, transporters, or metabolism...



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4.3 Ask one of the example questions by clicking it or type in your own question



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What are the inputs, regulators, and phenotypic...

Which triggers initiate the possible pathologic...

I have read and agree to the informed consent information and I want to participate in the Llemy user research.

☒ Yes ☐ No

 [View informed consent details](#)

Llemy

An Agentic System for MINERVA Map Exploration

☐ Retrieving knowledge from MINERVA project 'Liver_Lipid_Metabolism_Physiological_Map_August_2024' ...

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Ask about biology, transporters, or metabolism...

 Stop

4.4 The system can take some time answering

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4.5a Review the response and examine the links provided if any

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Which triggers initiate the possible pathologic...

Which regulatory checkpoints limit over-activat...

 What is the scope of this map? Give me a brief summary of the biology represented.



Scope of the Map and Biological Representation

The provided Minerva Map encompasses a broad spectrum of metabolic and signaling pathways primarily related to cellular energy metabolism, lipid and cholesterol homeostasis, amino acid metabolism, oxidative stress responses, and gene regulation mechanisms. The reactions involve key enzymes, transporters, modifiers, and signaling molecules that coordinate metabolic fluxes, lipid transport, lipoprotein metabolism, and mitochondrial function. The map also includes pathways for gluconeogenesis, glycolysis, fatty acid oxidation, ketogenesis, cholesterol biosynthesis and modification, and oxidative stress responses such as reactive oxygen species (ROS) generation and detoxification. Additionally, it features regulatory pathways involving transcription factors (e.g., SREBF1, PPARs, FOXO1), kinases (e.g., AKT1, GSK3B, mTORC1), and nuclear receptors (e.g., NR1H3/FXR, NR1I2/PXR), which modulate gene expression linked to metabolic processes.

Biological Themes Represented

- Lipid and Cholesterol Metabolism:** The map details reactions for cholesterol esterification, desmosterol to cholesterol conversion, bile acid synthesis, and lipoprotein metabolism (VLDL, LDL, HDL). Enzymes like DHCR24, DHCR7, CYP7A1, and EBP are involved in sterol biosynthesis and modification, indicating a focus on cholesterol homeostasis and steroid biosynthesis.
- Fatty Acid Metabolism:** It includes fatty acid synthesis, elongation (ELOVL family), desaturation (SCD), beta-oxidation in mitochondria and peroxisomes, and transport via fatty acyl-CoA and carnitine shuttles. Reactions involving ECHS1, ACADL, and CPT1A highlight mitochondrial fatty acid oxidation pathways. [\(link to MINERVA map\)](#)
- Amino Acid and Nitrogen Metabolism:** Reactions for amino acid catabolism (e.g., glutamate, serine, glycine, threonine), transamination, decarboxylation, and urea cycle components are present, reflecting nitrogen handling and amino acid interconversions.
- Energy Production and Mitochondrial Function:** The map includes mitochondrial electron transport chain components (complex I, II, III, IV), TCA cycle reactions (isocitrate dehydrogenase, alpha-ketoglutarate dehydrogenase), and mitochondrial DNA transcription/translation, emphasizing oxidative phosphorylation and mitochondrial biogenesis.
- Oxidative Stress and ROS Handling:** Reactions involving superoxide, hydrogen peroxide, and enzymes like SOD1, SOD3, PRDXs, and catalase demonstrate mechanisms for ROS detoxification and redox regulation. [\(link to MINERVA map\)](#)

➤

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4.5b If you have agreed to data being recorded, you will see a feedback section for the answer, please fill it in

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Overall, the Minerva Map provides a comprehensive overview of interconnected metabolic and signaling pathways central to cellular energy management, lipid and structural homeostasis, oxidative stress response, and gene regulation. It captures the complexity of metabolic fluxes, transport mechanisms, enzymatic modifications, and regulatory controls that sustain cellular and systemic metabolic health. The inclusion of mitochondrial processes, lipid transport, and transcriptional regulation underscores its focus on the integrated biology of metabolism and cellular signaling pathways.

Note on Data Retrieval

The data was successfully retrieved and parsed from the Minerva Map, with detailed reaction annotations supporting this synthesis. No errors were indicated in the data extraction process.

Please give feedback on this answer

Accuracy

Score (1 = Not Accurate at all - 5 = Very accurate):

☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5

Comment on accuracy:

What was inaccurate?

Conciseness

Score (1 = Very verbose - 5 = Very concise):

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4.5c Do not forget to submit your feedback

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Score (1 = Very verbose - 5 = Very concise):

☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5

Reliability

Score (1 = Not reliable at all - 5 = Very reliable):

☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5

Comment on reliability:

2

☒ Feedback saved, thank you!

> Minerva API Data Retrieved

> API Call Status Details

Ask about biology, transporters, or metabolism...

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4.6 Repeat 4.1 to 4.5 as desired with new questions

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Score (1 = Very verbose - 5 = Very concise):

☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5

Reliability

Score (1 = Not reliable at all - 5 = Very reliable):

☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5

Comment on reliability:

2

Submit Feedback

✓ Feedback saved, thank you!

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☒ Yes ☐ No



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You can still copy your ID and/or change your consent at any time at the top of the app